Each counter is a person. To play a circle of 10 people, put counters on spots 1 to 10 of a mat and leave the other spots empty. Start at 1 and go round and round the circle. The people standing take turns: 1 stays standing, the next one sits down, the next one stays standing, the next one sits down, and so on. When you come round past 1 again, keep taking turns from where you left off. Flip a counter over when that person sits down. Skip empty spots and people who are sitting. Stop when only one person is standing.

Problem 1: Play circles of 1 to 16 people on the mat with 16 spots. Write the last one standing for each circle.

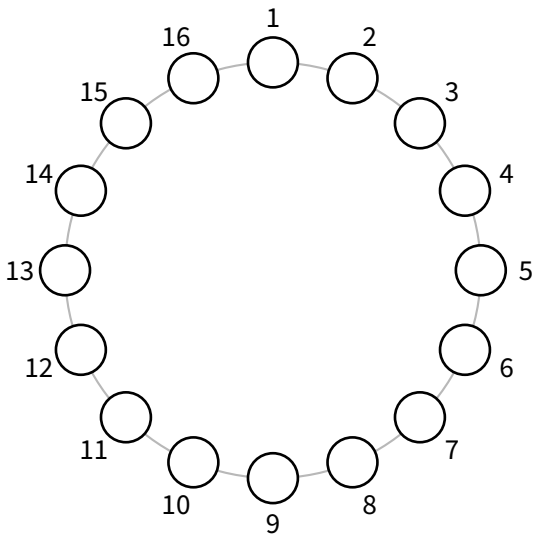
people	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**
last one standing								

people	**9**	**10**	**11**	**12**	**13**	**14**	**15**	**16**
last one standing								

Problem 2: Predict the last one standing in circles of 20, 25 and 32 people. Then play them on the mat with 32 spots to check.

people	**20**	**25**	**32**
prediction			
last one standing			

Problem 3: Find every circle up to 100 people where number 1 is the last one standing. Explain why number 1 is last in all of these circles.



Problem 4: Who is the last one standing in a circle of 100 people? In a circle of 1000 people? Explain why your answers are right.

Problem 5: Number 21 is the last one standing in a circle of 26 people. Explain how this tells you who is last in a circle of 52 people, and who is last in a circle of 53 people.

Problem 6: Find every circle up to 100 people where number 23 is the last one standing. Do the same for number 22. If there are none, explain why.

Number 23:

Number 22:

Problem 7: Now every third person sits down: 1 and 2 stay standing, the next one sits down, the next two stay standing, the next one sits down, and so on. Play circles of 1 to 12 people and write the last one standing for each.

people	**1**	**2**	**3**	**4**	**5**	**6**
last one standing						

people	**7**	**8**	**9**	**10**	**11**	**12**
last one standing						

Problem 8: With every third person sitting down, predict the last one standing in circles of 13 to 24 people. Then play circles of 14 and 21 to check.

people	**13**	**14**	**15**	**16**	**17**	**18**
prediction						
last one standing						

people	**19**	**20**	**21**	**22**	**23**	**24**
prediction						
last one standing						

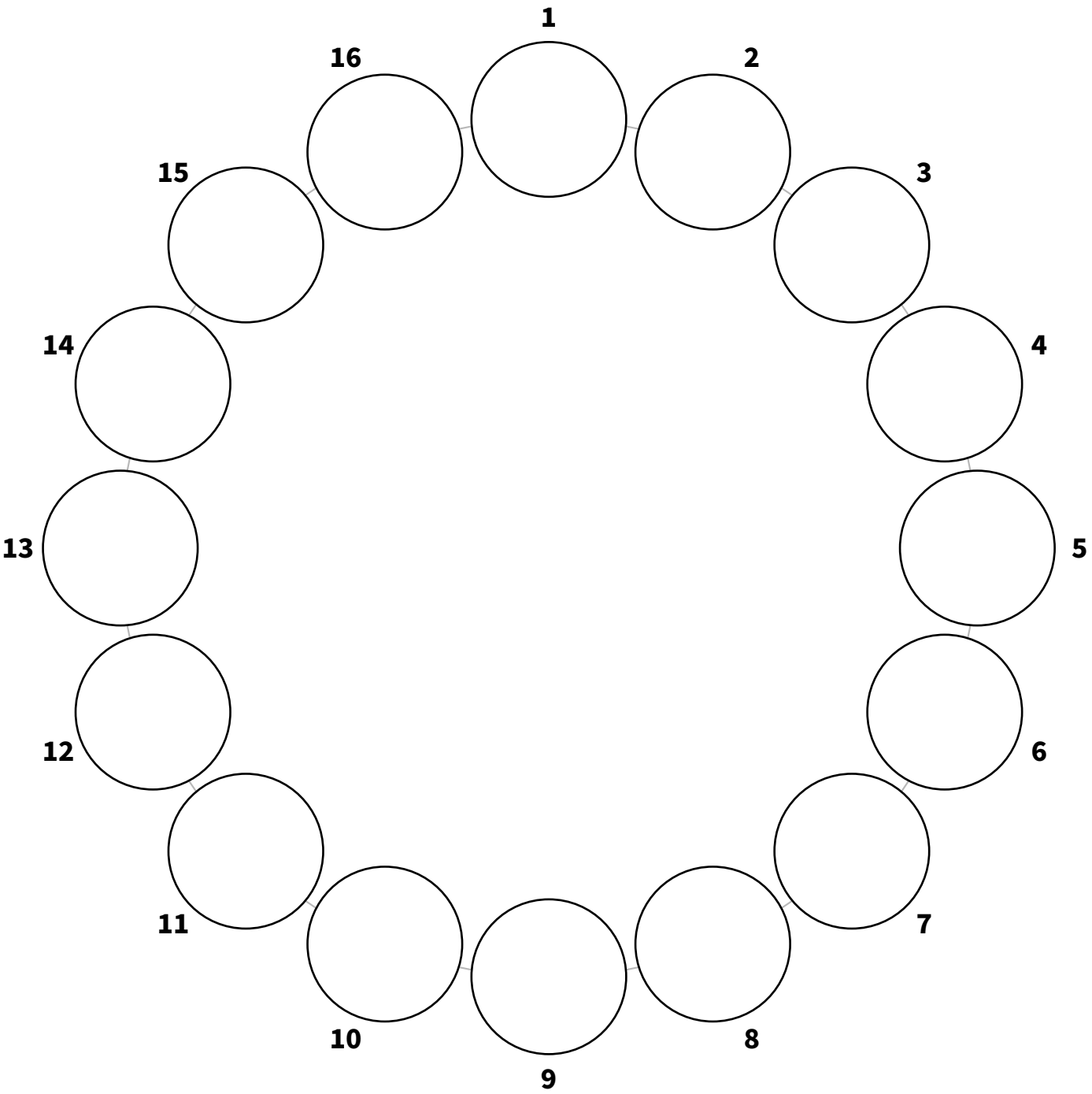
Problem 9: In a circle of 41 people, every third person sits down. Where should you stand to be the last one standing?

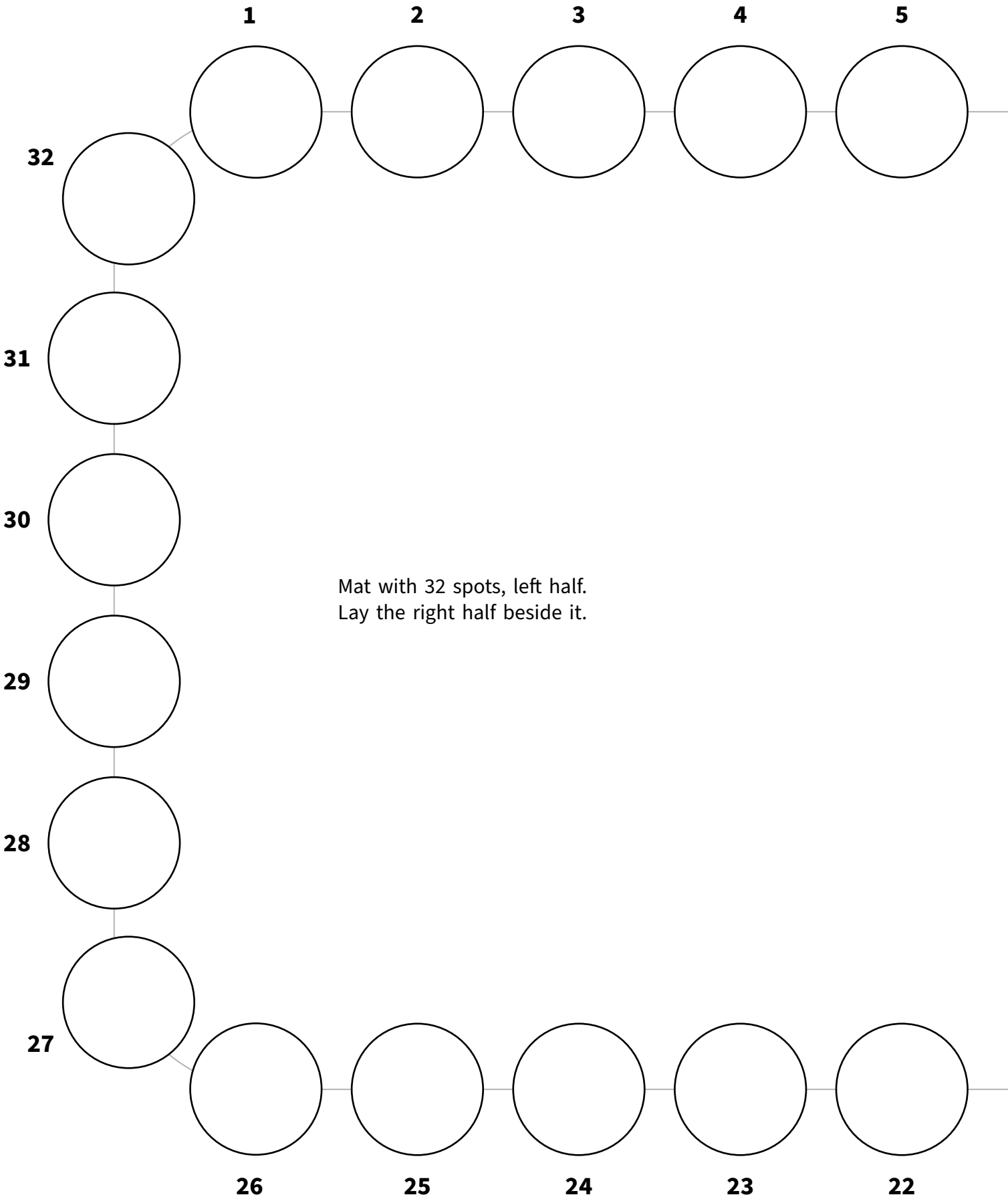
Problem 10: With every second person sitting down, the last one standing in a circle of 12 people is 9. In binary, 12 is written 1100, because 12 = 8 + 4, and 9 is written 1001, because 9 = 8 + 1. Find a rule that turns the number of people, written in binary, into the last one standing, written in binary. Explain why your rule works.

people	in binary					
	32	**16**	**8**	**4**	**2**	**1**
12			1	1	0	0

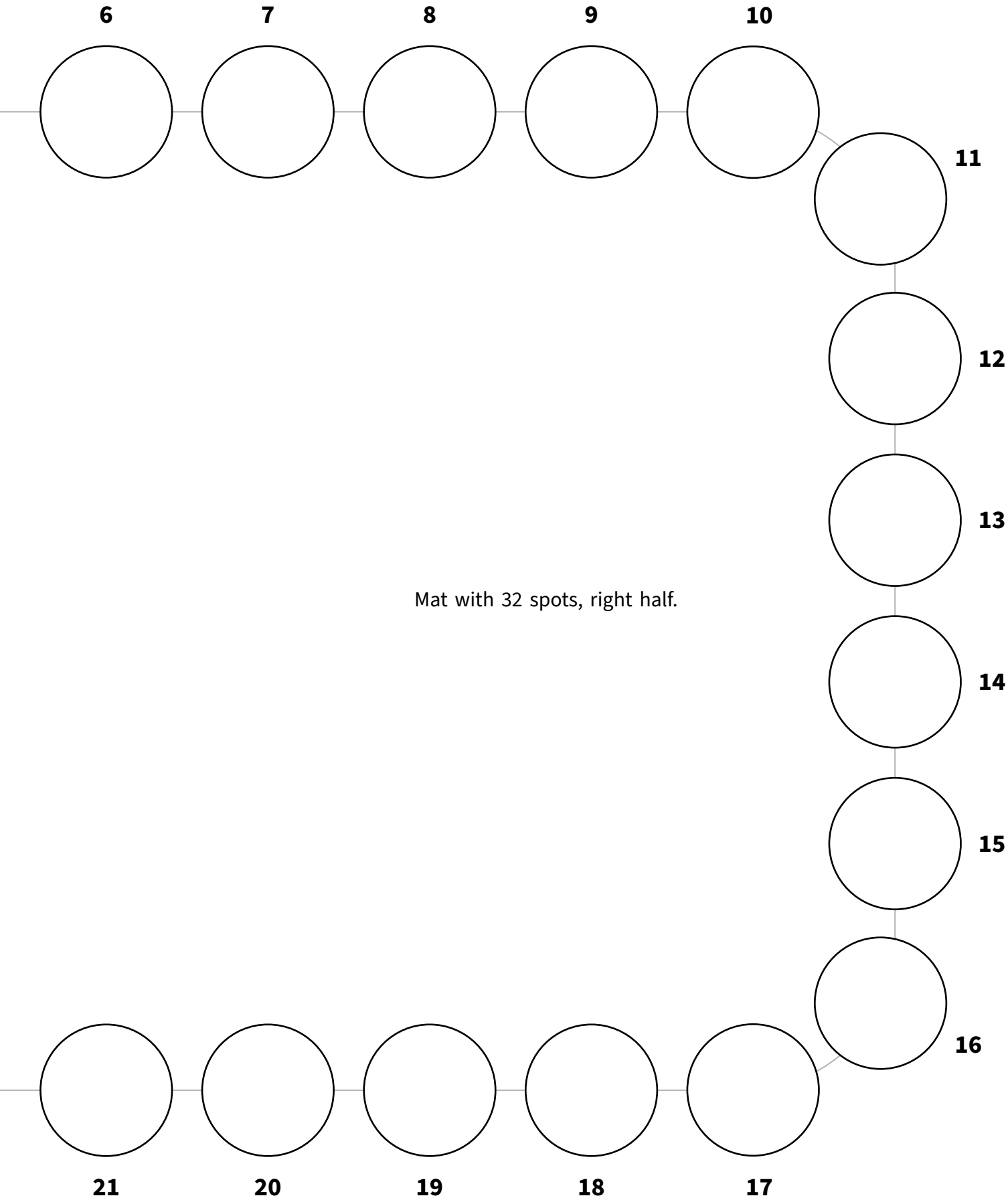
last one standing	in binary					
	32	**16**	**8**	**4**	**2**	**1**
9			1	0	0	1

Mat with 16 spots.





Mat with 32 spots, left half.
Lay the right half beside it.



Mat with 32 spots, right half.